

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**  
Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

**Course Code: MAT204**

**Course Name: PROBABILITY, RANDOM PROCESSES AND NUMERICAL METHODS**

Max. Marks: 100

Duration: 3 Hours

**(Statistical Tables are allowed)**

**PART A**

*(Answer all questions; each question carries 3 marks)*

|    |                                                                                                                                                                               | Marks |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1  | In a binomial distribution, if the mean is 6, and the variance is 4, find $P[X=1]$ .                                                                                          | 3     |
| 2  | Given that $f(x) = \frac{K}{2^x}$ is a probability mass function of a random variable that can take on the values $x = 0, 1, 2, 3$ and 4, find (i) K and (ii) $P(X \leq 2)$ . | 3     |
| 3  | Find the mean and variance for the PDF, $f(x) = \begin{cases} Kx^2, & 0 < X < 1 \\ 0, & \text{elsewhere} \end{cases}$                                                         | 3     |
| 4  | If random variable X has a uniform distribution in $(-3, 3)$ , find $P( X - 2  < 2)$ .                                                                                        | 3     |
| 5  | Define stationary random process. Define two types of stationary random process.                                                                                              | 3     |
| 6  | Write down the properties of the power spectral density.                                                                                                                      | 3     |
| 7  | Write down the Newton's forward and backward difference interpolation formula                                                                                                 | 3     |
| 8  | Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ with 4 subintervals by Simpson's rule.                                                                                                 | 3     |
| 9  | Write the normal equations for fitting a curve of the form $y = a + bx + cx^2$ to a given set of pairs of data points.                                                        | 3     |
| 10 | Using Euler's method, find y at $x = 0.25$ , given $y' = 2xy$ , $y(0) = 1$ , $h = 0.25$ .                                                                                     | 3     |

**PART B**

*(Answer one full question from each module, each question carries 14 marks)*

**Module -1**

- |    |    |                                                                                                                                                                                                                                      |   |
|----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 11 | a) | Six dice are thrown 729 times. How many times do you expect at least three dice to show 1 or 2?                                                                                                                                      | 6 |
|    | b) | Derive the formula for mean and variance of Poisson distribution                                                                                                                                                                     | 8 |
| 12 | a) | A random variable X takes the values -3, -2, -1, 0, 1, 2, 3 such that $P(X=0) = P(X>0) = P(X<0)$ and $P(X=-3) = P(X=-2) = P(X=-1) = P(X=1) = P(X=2) = P(X=3)$ . Obtain the probability mass function and distribution function of X. | 7 |

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- b) The joint probability distribution of X and Y is given by,  $f(x, y) = \frac{1}{27}(2x + y)$ ; 7  
 $x = 0, 1, 2$  and  $y = 0, 1, 2$ .  
(i) Find the marginal distributions of X and Y.  
(ii) Are X and Y independent random variables.

**Module -2**

- 13 a) Suppose the diameter at breast height (in.) of trees of a certain type is normally distributed with mean 8.8 and standard deviation 2.8. (i) What is the probability that the diameter of a randomly selected tree will be at least 10 in.? (ii) What is the probability that the diameter of a randomly selected tree will exceed 20 in.? (iii) What is the probability that the diameter of a randomly selected tree will be between 5 in. and 10 in.? 7
- b) The amount of time that a surveillance camera will run without having to be reset is a random variable having exponential distribution with mean 50 days. Find the probabilities that such a camera will (a) have to be reset in less than 20 days. (b) not have to be reset in at least 60 days. 7
- 14 a) The joint density function of 2 continuous random variable X and Y is 7  
$$f(x, y) = \begin{cases} cxy & ; 0 < x < 4, 1 < y < 5 \\ 0 & ; \text{otherwise} \end{cases}$$
  
(i) Find the value of the constant c.  
(ii) Find  $P(X \geq 3, Y \leq 2)$   
(iii) Find the marginal density of X.
- b) The life time of a certain brand of tube light may be considered as a random variable with mean 1200 hours and standard deviation 250 hours. Using Central limit theorem, find the probability that the average life time of 60 lights exceeds 1250. 7

**Module -3**

- 15 a) Let  $X(t) = A \cos \lambda t + B \sin \lambda t$ , where A and B are independent normally distributed random variables  $N(0, \sigma^2)$ . Show that  $X(t)$  is WSS. 7
- b) If  $X(t) = A \cos(\omega t + \theta)$  Where A and  $\omega$  are constants and  $\theta$  is uniformly distributed over  $[0, 2\pi]$ , find the auto correlation function and Power Spectral Density of the process. 7

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- 16 a) Assume that  $X(t)$  is a random process defined as follows:  $X(t) = A \cos(2\pi t + \phi)$  7  
 where  $A$  is a zero-mean normal random variable with variance  $\sigma_A^2 = 2$  and  $\phi$  is uniformly distributed random variable over the interval  $-\pi \leq \phi \leq \pi$ .  $A$  and  $\phi$  are statistically independent. Let the random variable  $Y$  be defined as  $Y = \int_0^1 X(t) dt$ . Determine (i) The mean of  $Y$  (ii) The variance of  $Y$ .
- b) If the customers arrive at a counter in accordance with Poisson distribution with rate of 2 per minute. Find the probability that the interval between two consecutive arrivals is (i) more than 1 minute (ii) between 1 minute and 2 minutes. 7

**Module -4**

- 17 a) Use Newton- Raphson method to find a non- zero solution of  $f(x)=2x - \cos x = 0$  7  
 b) Using Lagrange's interpolating polynomial estimate  $y(5)$  for the following data: 7

|   |    |   |    |     |
|---|----|---|----|-----|
| x | 1  | 3 | 4  | 6   |
| y | -3 | 0 | 30 | 132 |

- 18 a) Find the polynomial interpolating the following data, using Newton's backward interpolating formula 7

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| x | 3 | 4  | 5  | 6  | 7  |
| y | 7 | 11 | 16 | 22 | 29 |

- b) Using Newton's divided difference formula, evaluate  $y(8)$  and  $y(15)$  from the following data 7

|   |    |     |     |     |      |      |
|---|----|-----|-----|-----|------|------|
| x | 4  | 5   | 7   | 10  | 11   | 13   |
| y | 48 | 100 | 294 | 900 | 1210 | 2028 |

**Module -5**

- 19 a) Solve the following system of equations using Gauss- Seidel iteration method starting with the initial approximation  $(0,0,0)^T$  7

$$8x_1 + x_2 + x_3 = 8$$

$$2x_1 + 4x_2 + x_3 = 4$$

$$x_1 + 3x_2 + 5x_3 = 5$$

- b) Fit a straight-line  $y = ax + b$  for the following data: 7

|   |   |   |   |   |   |   |    |    |
|---|---|---|---|---|---|---|----|----|
| X | 1 | 3 | 4 | 6 | 8 | 9 | 11 | 14 |
| Y | 1 | 2 | 4 | 4 | 5 | 7 | 8  | 9  |

- 20 a) Solve the following system of equations using Gauss- Jacobi iteration method 7  
starting with the initial approximation  $(0,0,0)^T$

$$20x_1 + x_2 - 2x_3 = 17$$

$$3x_1 + 20x_2 - x_3 = -18$$

$$2x_1 - 3x_2 + 20x_3 = 25$$

- b) Use Runge - Kutta method of fourth order to find  $y(0.1)$  from  $\frac{dy}{dx} = \sqrt{x+y}$ , 7  
 $y(0)=1$  taking  $h=0.1$ .

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